

PROBABILITY AND STATISTICS

UNIT 1 – PROBABILITY

Definitions, theorems, worked examples, and question bank

Exam Focus.

Unit scope. Sample spaces, events, approaches to probability, addition and multiplication theorems, conditional probability, independent and dependent events, odds, and Bayes theorem.

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1. Probability, sample spaces, and events

1.1 Probability

Probability is a numerical measure of how likely an event is to occur. For every event A ,

$$0 \leq P(A) \leq 1.$$

$P(A) = 0$ means that A is impossible; $P(A) = 1$ means that A is certain.

1.2 Random experiment and sample space

Term	Meaning	Example
Random experiment	A repeatable process whose exact outcome cannot be predicted with certainty.	Tossing a coin; drawing a card.
Outcome	One possible result of the experiment.	H in one coin toss.
Sample space S	The set of all possible outcomes.	$S = \{H, T\}$.
Event	One or more outcomes selected from S .	$A = \{H\}$ or “a king”.
Favourable outcomes	Outcomes that make the event occur.	For an even die result: $\{2, 4, 6\}$.

1.3 Types of sample space

- **Finite:** a limited number of outcomes, for example $\{1, 2, 3, 4, 5, 6\}$.
- **Countably infinite:** outcomes can be listed one by one, for example $\{0, 1, 2, 3, \dots\}$.
- **Continuous / uncountable:** outcomes form a continuum, for example a waiting time in an interval.
- A discrete sample space is finite or countably infinite; a continuous sample space is uncountably infinite.

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Example. For one die, $S = \{1, 2, 3, 4, 5, 6\}$. The event “even number” is $E = \{2, 4, 6\}$ and the event “number greater than 2” is $G = \{3, 4, 5, 6\}$.

1.4 Event notation

Notation	Meaning	Key fact
A^c	Complement of A ; A does not occur.	$P(A^c) = 1 - P(A)$.
$A \cup B$	A or B ; at least one occurs.	Union.
$A \cap B$	A and B ; both occur.	Intersection.
$\emptyset$	Null / impossible event.	$P(\emptyset) = 0$.
S	Certain event / sample space.	$P(S) = 1$.

Remember.

For “at least one”, use $P(\text{at least one}) = 1 - P(\text{none})$ when the complement is easier to calculate.

1.5 Types of events

Type	Definition	Example
Simple	Contains one outcome.	$\{5\}$ on a die.
Compound	Contains more than one outcome.	$\{2, 3, 5\}$.
Mutually exclusive	Cannot occur together; $A \cap B = \emptyset$.	Even and odd on one die.
Exhaustive	Events together cover S .	$\{H\}$ and $\{T\}$ for one toss.
Equally likely	Outcomes have equal probability.	Heads and tails for an ideal coin.

2. Approaches to probability

2.1 Classical approach

When outcomes are equally likely,

$$P(A) = \frac{\text{number of favourable outcomes}}{\text{total number of equally likely outcomes}}.$$

For a king from a standard deck, $P(\text{king}) = 4/52 = 1/13$. For two simultaneous coin tosses, $S = \{HH, HT, TH, TT\}$, so $P(\text{one head and one tail}) = 2/4 = 1/2$.

2.2 Relative-frequency approach

If an event occurs r times in n suitable trials, its observed relative frequency is r/n . This is an estimate based on data, not automatically an exact probability.

2.3 Subjective probability

Subjective probability is based on judgement, information, or past experience, for example a weather forecast or a medical assessment.

2.4 Axiomatic approach

- $P(A) \geq 0$ for every event A .
- $P(S) = 1$.
- If A and B are mutually exclusive, $P(A \cup B) = P(A) + P(B)$.

Do Not Confuse.

The source deck labels the conditional-probability formula as applying to independent events. The formula is general: $P(A | B) = P(A \cap B)/P(B)$ whenever $P(B) > 0$. Independence is an additional condition, not a requirement for defining conditional probability.

3. Counting outcomes: dice and tables

For two fair dice, ordered outcomes are pairs (i, j) and there are $6 \times 6 = 36$ equally likely outcomes. The order matters: $(2, 5)$ and $(5, 2)$ are different outcomes.

Sum	Favourable ordered pairs	Count	Probability
2	(1, 1)	1	1/36
3	(1, 2), (2, 1)	2	2/36
4	(1, 3), (2, 2), (3, 1)	3	3/36
5	Four pairs	4	4/36
6	Five pairs	5	5/36
7	Six pairs	6	6/36 = 1/6
8	Five pairs	5	5/36
9	Four pairs	4	4/36
10	Three pairs	3	3/36
11	Two pairs	2	2/36
12	(6, 6)	1	1/36

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Two-dice example. $P(\text{sum} = 8) = 5/36$, $P(\text{first die} = 6) = 6/36 = 1/6$, $P(\text{sum} > 8) = 10/36 = 5/18$, $P(\text{sum} = 13) = 0$, $P(\text{same number}) = 6/36 = 1/6$, $P(\text{sum} < 5) = 6/36 = 1/6$, and $P(\text{sum} = 6) = 5/36$.

4. Addition theorem

4.1 Non-mutually-exclusive events

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

The intersection is subtracted because it is counted twice in $P(A) + P(B)$.

Exam Focus.

Example. A student passes Physics with probability $2/3$, passes both Physics and English with probability $14/45$, and passes at least one with probability $4/5$. If E is passing English,

$$\frac{4}{5} = \frac{2}{3} + P(E) - \frac{14}{45} \Rightarrow P(E) = \frac{4}{9}.$$

4.2 Mutually exclusive events

If $A \cap B = \emptyset$, then

$$P(A \cup B) = P(A) + P(B).$$

For 4 white, 3 black, and 5 red balls, $P(\text{white or red}) = (4 + 5)/12 = 3/4$.

Do Not Confuse.

Do not use $P(A \cup B) = P(A) + P(B)$ when the events overlap. Subtract $P(A \cap B)$ unless mutual exclusivity is given or proved.

5. Conditional probability and multiplication

5.1 Conditional probability

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0.$$

The notation $A | B$ means “ A given that B has occurred.”

5.2 Multiplication theorem

$$P(A \cap B) = P(A)P(B | A) = P(B)P(A | B).$$

If A and B are independent,

$$P(A \cap B) = P(A)P(B).$$

Point	Independent	Dependent
Does the first event change the next probability?	No.	Yes.
Typical setup	Separate tosses; replacement.	Without replacement.
Formula	$P(A \cap B) = P(A)P(B)$.	$P(A \cap B) = P(A)P(B A)$.

5.3 Worked examples

Exam Focus.

Two black cards without replacement.

$$P(\text{both black}) = \frac{26}{52} \times \frac{25}{51} = \frac{25}{102} \approx 0.2451.$$

The second denominator is 51 because the first card is not replaced.

Exam Focus.

At least one student solves. Three students have probabilities $1/2$, $1/3$, and $1/4$ of solving, independently. Then

$$P(\text{none}) = \frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} = \frac{1}{4}, \quad P(\text{at least one}) = 1 - \frac{1}{4} = \frac{3}{4}.$$

Exam Focus.

Survival. If $P(X \text{ alive}) = 0.8$ and $P(Y \text{ alive}) = 0.6$, assuming independence: both alive = 0.48; neither alive = 0.08; either alive = $1 - 0.08 = 0.92$.

6. Probability trees and odds

6.1 Probability tree

For two black cards without replacement:

$$\begin{array}{l} \text{First black } (26/52) \longrightarrow \text{Second black } (25/51) \\ \text{First red } (26/52) \longrightarrow \text{Second black } (26/51) \end{array}$$

Multiply probabilities along one path for “and”. Add separate paths for “or”.

6.2 Odds

If odds in favour of A are $a : b$, then

$$P(A) = \frac{a}{a+b}.$$

If odds against A are $a : b$, then $P(A) = b/(a+b)$. Thus odds in favour of winning 9 : 2 give $P(\text{win}) = 9/11$.

Exam Focus.

Contradiction example. If $P(A \text{ truthful}) = 4/5$ and $P(B \text{ truthful}) = 3/4$, and the truthfulness events are independent, contradiction means exactly one tells the truth:

$$\frac{4}{5} \frac{1}{4} + \frac{1}{5} \frac{3}{4} = \frac{7}{20}.$$

7. Bayes theorem

Bayes theorem finds the probability of a cause after observing evidence:

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}.$$

For mutually exclusive and exhaustive causes $A_1, \dots, A_n$,

$$P(A_i | B) = \frac{P(B | A_i)P(A_i)}{\sum_{j=1}^n P(B | A_j)P(A_j)}.$$

7.1 Bayes table

Cause	Prior $P(A_i)$	Evidence $P(B A_i)$	Product
Company A	0.40	0.05	0.0200
Company B	0.30	0.03	0.0090
Company C	0.30	0.025	0.0075
Total			0.0365

7.2 Worked example: late computer

If a computer is late, the probability it came from Company A is

$$P(A | L) = \frac{0.40(0.05)}{0.40(0.05) + 0.30(0.03) + 0.30(0.025)} = \frac{0.0200}{0.0365} \approx 0.5479.$$

7.3 Worked example: urns

Urn 1 has 6 black and 4 red balls; Urn 2 has 2 black and 2 red balls. Select an urn at random, then draw one ball.

$$P(B) = \frac{1}{2} \frac{6}{10} + \frac{1}{2} \frac{2}{4} = \frac{11}{20}, \quad P(U_1 | B) = \frac{(1/2)(6/10)}{11/20} = \frac{6}{11}.$$

7.4 Common Bayes errors

- Reversing $P(A | B)$ and $P(B | A)$.
- Omitting the prior probability.
- Using only the target row instead of the total evidence probability.
- Assuming a test is fully specified by the word “accurate.” Sensitivity and false-positive rate must be stated.

8. Important exam topics

- Define random experiment, outcome, sample space, event, and favourable outcome.
- Classify events: simple, compound, mutually exclusive, exhaustive, complementary, independent, and dependent.
- Apply the classical formula to cards, dice, coins, and urns.
- Use the addition theorem for overlapping and mutually exclusive events.
- Use conditional probability and the multiplication theorem.
- Distinguish independence from mutual exclusivity.
- Convert odds to probability.
- Construct a Bayes table and identify posterior probability.

9. Question bank

Definitions / 2–3 marks

1. Define probability and state its range.
2. Define random experiment, sample space, outcome, and event.
3. Distinguish simple and compound events.
4. Define mutually exclusive and exhaustive events.
5. State the complement rule and the addition theorem.
6. Define conditional probability and independent events.
7. State Bayes theorem.

5-mark questions

1. Two dice are thrown. Find the probability that the sum is 7, the first die is 4, the sum exceeds 6, the sum is 12, the dice differ, the sum is less than 6, and the sum is 8.
2. A committee of three is chosen from 5 men and 5 women. Find the probability of all men and of exactly two men.
3. A key is in a bunch of 10 keys; wrong keys are discarded. Find the probability that the door opens on the fifth trial.
4. $P(A) = 0.25$, $P(B) = 0.40$, and $P(A \cap B) = 0.15$. Find the probability that neither occurs.
5. A and B speak truth with probabilities $4/5$ and $3/4$. Find the probability that they contradict each other.

Bayes theorem questions

- Three companies supply 40%, 30%, and 30% of computers and are late 5%, 3%, and 2.5% of the time. Given a late computer, find the probability it came from Company A.
- Adults are 51% male; 9.5% of males and 1.7% of females smoke cigars. Given a cigar smoker, find the probability of male.
- A person drives 30%, walks 30%, and takes the bus 40% of the time. Late rates are 3%, 10%, and 7%. Find $P(\text{bus} \mid \text{late})$ and $P(\text{walk} \mid \text{on time})$.
- Four boxes contain respectively 1, 2, 3, and 5 \$100 bills out of 10. A box is selected at random and a \$100 bill appears. Find the probability it came from the box with five \$100 bills.
- Three factories produce 50%, 30%, and 20% of a model; white-car rates are 30%, 40%, and 25%. Find $P(\text{white})$ and $P(B \mid \text{white})$.

10. Answer key

Question	Answer
Two dice: sum 7; first die 4; sum > 6; sum 12; different; sum < 6; sum 8	1/6, 1/6, 7/12, 1/36, 5/6, 5/18, 5/36.
Survival	0.48 for both alive, assuming independence.
Leap year with 53 Sundays	5/7, assuming starting weekdays are equally likely.
Three students	3/4.
Committee: all men; two men	1/12; 5/12.
Fifth key trial	1/10.
Neither A nor B	0.50.
Contradiction	7/20.
Late computer from A	$0.0200/0.0365 \approx 0.5479$.
Bayes: cigar smoker is male	Approximately 0.853.
Bus given late; walk given on time	Approximately 0.424; 0.290.
Yellow box given \$100 bill	4/11.
AirCorp given defective	Approximately 0.7033.
Biased coin given three heads	$27/43 \approx 0.628$.
White car; Factory B given white	0.32; 0.375.

11. Last-minute revision page

CORE CONCEPT

Probability measures how likely an event is; $0 \leq P(A) \leq 1$.

KEY TERMS

Experiment: process; outcome: result; sample space: all results; event: selected results.

COMPLEMENT

$$P(A^c) = 1 - P(A).$$

ADDITION

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

CONDITIONAL

$$P(A \mid B) = P(A \cap B)/P(B).$$

MULTIPLICATION

$$P(A \cap B) = P(A)P(B | A).$$

INDEPENDENCE

$$P(A \cap B) = P(A)P(B).$$

BAYES

Posterior = likelihood $\times$ prior / evidence.

ODDS

Odds $a : b$ in favour gives $P = a/(a + b)$.

DO NOT CONFUSE

“And” means intersection; “or” means union. Mutual exclusivity is not the same as independence. $P(A | B)$ is not generally $P(B | A)$.

BAYES TABLE

List every cause, prior, conditional evidence probability, product, and total.

Source scope and precision notes

These notes are based on *Module 1.pdf*, *Practise – Probability.pdf*, and *Practise Bayes Theorem.pdf*. The organization follows the supplied module. The source statement that conditional probability is only for independent events has been corrected; the source example that multiplies marginal probabilities for selecting a Hindi-knowing woman teacher requires an independence assumption that is not established by the given counts.